

Singular polarizations

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From proof of Helffer-Nourrigat conjecture

- ▶ Filtration of f.g. $C^\infty(M)$ -submodules:

$$\mathcal{F}^\bullet : \quad \mathcal{F}^1 \subset \mathcal{F}^2 \subset \dots \subset \mathcal{F}^N = \mathcal{X}(M), \quad [\mathcal{F}^i, \mathcal{F}^j] \subset \mathcal{F}^{i+j}$$

- ▶ (Singular) foliation on $M \times \mathbb{R}$:

$$\mathcal{F} = t\mathcal{F}^1 + t^2\mathcal{F}^2 + \dots + t^N\mathcal{F}^N$$

- ▶ Nilpotent Lie algebra at $x \in M$:

$$\mathfrak{g}_x = \frac{\mathcal{F}}{I_{(x,0)}\mathcal{F}} = \frac{\text{gr}(\mathcal{F}^\bullet)}{I_x \text{gr}(\mathcal{F}^\bullet)}$$

- ▶ Noncommutative deformation groupoid:

$$H(\mathcal{F}) = (\text{Gr}(\mathcal{F}^\bullet) \times \{0\}) \coprod (M \times M \times \mathbb{R}^*) \rightrightarrows M \times \mathbb{R}$$

- ▶ (Equi)regular case:

$$0 \rightarrow K(L^2(M)) \otimes \mathbb{R}^* \rightarrow C^*(\mathcal{F}) \rightarrow C_0(\widehat{\text{Gr}(\mathcal{F}^\bullet)}) \rightarrow 0$$

Weinstein's GQ scheme for Poisson (M, π)

- ▶ Say Lie algebroid $\pi^\sharp : T^*M \rightarrow TM$ integrates to symplectic groupoid $(\Sigma, \omega_\Sigma) \rightrightarrows M$.
- ▶ Idea: Perform GQ on (Σ, ω_Σ) ; then “push” down to M .

Hawkins 2006: Gpd approach to GQ of Poisson mfd (M, π)

- ▶ Symplectic gpd $(\Sigma, \omega) \implies M$
- ▶ Hermitian line bundle $L \rightarrow \Sigma$ with $\text{curv}(L) = \omega$,

$$\sigma(\eta_1, \eta_2) : L_{\eta_1} \otimes L_{\eta_2} \rightarrow L_{\eta_1 \eta_2} \quad \text{bilinear}$$

- ▶ Twisted half-densities: $C_c^\infty(\Sigma; L \otimes \Omega^{1/2}\Sigma)$:

$$(a * b)(\gamma) = \int_{\eta_1 \eta_2 = \gamma} \sigma(\eta_1, \eta_2) a(\eta_1) b(\eta_2)$$

Get $C^*(\Sigma, \sigma)$.

- ▶ Polarization: Lagrangian LA-subgroupoid P of $T\Sigma \implies TM$.
- ▶ Twisting by polarised densities: Assume

$$\Sigma/P \implies M/P_M \text{ Lie groupoid, } L = p^* L_0 \quad (p : \Sigma \rightarrow \Sigma/P)$$

Polarised sections:

$$C_P^*(\Sigma, \sigma) := C^*(\Sigma/P, \sigma_0)$$

Issues with Hawkins' approach

- 1 No quantization map. Hunch: Use DNC groupoid

$$T^*M \times \{0\} \coprod \Sigma \times \mathbb{R}^\times \Longrightarrow M \times \mathbb{R}$$

Rieffel, Landsman, Ramazan: *-product for Lie-Poisson structures.

- 2 Σ/P , M/P_M not smooth manifolds!

- 3 How to push down to M ?

Weinstein says: Better deform to $(M \times \mathbb{R}, t\pi)$ first...

The deformation groupoid of a Poisson mfd (M, π)

1 Extension of Lie algebroids over M :

$$0 \rightarrow M \times \mathbb{R} \rightarrow T^*M \oplus_{\pi} (M \times \mathbb{R}) \rightarrow T^*_{\pi}M \rightarrow 0$$

- ▶ anchor: $\rho(\alpha \oplus f) = \pi(\alpha)$
- ▶ bracket: $[(\alpha \oplus f), (\beta \oplus g)]'_{\pi} = [\alpha, \beta]_{\pi} \oplus \pi(\alpha \wedge \beta)$

2 Filtration:

$$\pi^{\bullet} : \quad T^*M \subset T^*M \oplus_{\pi} (M \times \mathbb{R})$$

3 Deformation algebroid:

$$\mathcal{D} = t\Omega^1(M) + t^2(\Omega^1(M) \oplus C^{\infty}(M))$$

- ▶ anchor: $\rho(t\alpha) = t\pi(\alpha)$, $\rho(t^2f) = 0$
- ▶ bracket: $[t\alpha, t\beta] = t^2[\alpha, \beta]'_{\pi} = t^2[\alpha, \beta]_{\pi} \oplus t^2\langle \pi, \alpha \wedge \beta \rangle$

4 Isomorphism of Lie algebroids:

$$\phi : \Gamma(T^*_{t\pi}(M \times \mathbb{R})) \rightarrow \mathcal{D}, \quad \phi(\alpha, f) = t\alpha + t^2f$$

The (symplectic) groupoid of $(M \times \mathbb{R}, t\pi)$

$$0 \rightarrow M \times \mathbb{R} \rightarrow T^*M \oplus_{\pi} (M \times \mathbb{R}) \rightarrow T_{\pi}^*M \rightarrow 0 \quad (1)$$

- ▶ Let $(\Sigma, \omega_{\Sigma}) \rightrightarrows M$ integration of T_{π}^*M .
- ▶ If ω_{Σ} integral, get S^1 -bundle $p : S \rightarrow \Sigma$ with

$$-d\eta = p^*\omega_{\Sigma}, \quad \eta \in \Omega^1(S) \text{ connection form}$$

- ▶ **contact** groupoid $(S, \eta) \rightrightarrows M$. Integration of (1):

$$1 \rightarrow M \times S^1 \rightarrow (S, \eta) \xrightarrow{p} (\Sigma, \omega_{\Sigma}) \rightarrow 1$$

- ▶ Then Heisenberg Poisson structure $(M \times \mathbb{R}, t\pi)$ integrates to deformation groupoid:

$$\Delta = (\text{Gr}(\pi^{\bullet}) \times \{0\}) \sqcup (S \times \mathbb{R}^{\times}) \rightrightarrows M \times \mathbb{R}$$

Osculating Symplectic structure of Δ

- ▶ $\mathcal{X}_c(\Delta) \simeq (\mathfrak{t}p^{-1}\mathcal{X}_c(\Sigma) + \mathfrak{t}^2\mathcal{X}_c(S)) \oplus \langle \partial\mathfrak{t} \rangle$
- ▶ $\Omega_c^1(\Delta) \simeq \left(\frac{1}{\mathfrak{t}}\Omega_c^1(S) + \frac{1}{\mathfrak{t}^2}\text{Ann}(p^{-1}\mathcal{X}_c(\Sigma))\right) \oplus \langle d\mathfrak{t} \rangle$
- ▶ $\tilde{\eta} = \frac{1}{\mathfrak{t}^2}p^*\eta$ lives in $\frac{1}{\mathfrak{t}^2}\text{Ann}(p^{-1}\mathcal{X}_c(\Sigma)) \subseteq \Omega_c^1(\Delta)$

Osculating symplectic form

$$\Omega_\Delta := -d(\mathfrak{t}\tilde{\eta}) = -\frac{1}{\mathfrak{t}}p^*\omega_\Sigma + \frac{1}{\mathfrak{t}^2}d\mathfrak{t} \wedge \eta$$

Remark

Ω_Δ exact, so (Δ, Ω_Δ) prequantisable by $L = \Delta \times \mathbb{C}$.

Connection: $d + \tilde{\eta}$.

Example: Symplectic sphere (S^2, ω) , spherical coordinates:

- ▶ $\omega_{(\theta, \phi)} = \sin(\phi) d\theta \wedge d\phi$, $\phi \in (0, \pi)$, $\theta \in (0, 2\pi)$.
- ▶ Prequantization: $S^3(S^2, S^1)$.
- ▶ Symplectic gpd: $S^2 \times S^2 \rightrightarrows S^2$, $\omega_\Sigma = \text{pr}_2^* \omega + \text{pr}_1^* \overline{\omega}$.
- ▶ Prequantizing bundle:

$$1 \rightarrow S^2 \times S^1 \rightarrow \frac{S^3 \times S^3}{S^1} \xrightarrow{p} S^2 \times S^2 \rightarrow 1$$

- ▶ Polarization by $\mathbb{T}^2$ action:

$$\mathcal{P}_{S^2} = \langle \partial_\phi \rangle, \quad \mathcal{P} = \mathcal{C} \oplus \mathcal{P}_{S^2} = \langle (\partial_\phi, 0), (0, \partial_\phi) \rangle, \quad \mathcal{C} = \langle (\partial_\phi, 0) \rangle$$

Singular: ∂_ϕ vanishes on 0 and π .

- ▶ $\text{gr}(\omega_\bullet) = \frac{TS^2 \oplus \mathbb{R}_{S^2}}{TS^2} \oplus TS^2 \longrightarrow S^2$

$$\Delta = (\text{Gr}(\omega) \times \{0\}) \sqcup \left(\left(\frac{S^3 \times S^3}{S^1} \right) \times \mathbb{R}^\times \right) \rightrightarrows S^2 \times \mathbb{R}$$

Polarization 1

Definition

Polarization of symplectic gpd $(\Sigma, \omega_\Sigma) \implies (M, \pi)$ is a pair $(\mathcal{P}, \mathcal{P}_M)$:

- 1 $\mathcal{P} \subset \mathcal{X}_c(\Sigma)$ Lagrangian foliation of (Σ, ω_Σ) ;
- 2 $\mathcal{P}_M \subset \mathcal{X}_c(M)$ coisotropic foliation of (M, π) ;
- 3 Double groupoid:

$$\begin{array}{ccc} H(\mathcal{P}) & \rightrightarrows & H(\mathcal{P}_M) \\ \downarrow \downarrow & & \downarrow \downarrow \\ \Sigma & \rightrightarrows & M \end{array} \quad (2)$$

Hopefully, $\mathcal{P}, \mathcal{P}_M$ are almost regular foliations, *i.e.* $H(\mathcal{P}), H(\mathcal{P}_M)$ are Lie groupoids. (Not always!)

So, there are Lie algebroids $\mathcal{P} \rightarrow \Sigma$ and $\mathcal{P}_M \rightarrow M$.

Polarization 2

Differentiating vertically the double groupoid gives VB-groupoid:

$$\begin{array}{ccc}
 P & \xrightarrow{\pi} & \Sigma \\
 \downarrow \downarrow & & \downarrow \downarrow \\
 P_M & \xrightarrow{\pi_M} & M
 \end{array} \tag{3}$$

Let $C \rightarrow M$ be the **core** of (3), namely

$$0 \longrightarrow \mathbf{r}_\Sigma^! C \longrightarrow P \longrightarrow \mathbf{s}_\Sigma^! P_M \longrightarrow 0$$

Whence:

$$P = \mathbf{r}_\Sigma^! C \oplus \mathbf{s}_\Sigma^! P_M \quad \text{and} \quad \mathcal{P} = \mathbf{r}_\Sigma^{-1} \mathcal{C} \oplus \mathbf{s}_\Sigma^{-1} \mathcal{P}_M \tag{4}$$

Osculation of polarization

Canonical surjective morphism of $C^\infty(M \times \mathbb{R})$ -modules

$$t\Gamma_c(A\Sigma) + t^2\Gamma_c(AS) \xrightarrow{q} t\Gamma_c(A\Sigma)$$

It is morphism of Lie algebroids $A\Delta \xrightarrow{q} A\Sigma_{\text{ad}}$, where

$$\Sigma_{\text{ad}} = \text{DNC}(\Sigma, M) \implies M \times \mathbb{R}$$

Whence, integrates to morphism of Lie groupoids

$$q : \Delta \rightarrow \Sigma_{\text{ad}}$$

Question: Polarization of $(\Delta, \Omega_\Delta) \implies (M \times \mathbb{R}, t\pi)$?

Answer: “Osculate” (P, P_M) :

- 1 Deform (P, P_M) to foliations $(\Sigma_{\text{ad}}, P_{\text{ad}})$, $(M \times \mathbb{R}, \text{pr}_1^{-1}(P_M))$;
- 2 Pull back to $\mathcal{Q} = q^{-1}(P_{\text{ad}})$.
- 3 Prove $(\mathcal{Q}, \text{pr}_1^{-1}(P_M))$ polarization of $(\Delta, \Omega_\Delta) \implies (M \times \mathbb{R}, t\pi)$.

Step 1: Deformation of $(\mathcal{P}, \mathcal{P}_M)$

Functoriality of DNC gives double Lie groupoid:

$$\begin{array}{ccc}
 \text{DNC}(\mathcal{H}(\mathcal{P}), \mathcal{H}(\mathcal{P}_M)) & \rightrightarrows & \mathcal{H}(\mathcal{P}_M) \times \mathbb{R} \\
 \downarrow \downarrow & & \downarrow \downarrow \\
 \Sigma_{\text{ad}} & \rightrightarrows & M \times \mathbb{R}
 \end{array} \tag{5}$$

Differentiating vertically:

$$A^V \text{DNC}(\mathcal{H}(\mathcal{P}), \mathcal{H}(\mathcal{P}_M)) = \text{DNC}(A^V \mathcal{H}(\mathcal{P}), A\mathcal{H}(\mathcal{P}_M)) = \text{DNC}(\mathcal{P}, \mathcal{P}_M) \rightarrow \Sigma_{\text{ad}}$$

Adiabatic deformation of $(\mathcal{P}, \mathcal{P}_M)$

$$\mathcal{P}_{\text{ad}} := A(\text{DNC}(\mathcal{P}, \mathcal{P}_M)) \quad \text{hence} \quad \mathcal{P}_{\text{ad}} = t\mathcal{P}$$

Lemma: $ts_{\Sigma}^{-1}(\mathcal{P}_M) \simeq s_{\Sigma_{\text{ad}}}^{-1}(\text{pr}_1^{-1}(\mathcal{P}_M))$ (as $C^\infty(\Sigma)$ -modules).

Calculation

$$\mathcal{P}_{\text{ad}} = (ts_{\Sigma}^{-1}\mathcal{P}_M) \oplus (t\mathbf{r}_{\Sigma}^{-1}\mathcal{C}) = \left(s_{\Sigma_{\text{ad}}}^{-1}(\text{pr}_1^{-1}\mathcal{P}_M)\right) \oplus (t\mathbf{r}_{\Sigma}^{-1}\mathcal{C}) \tag{6}$$

Step 2: Osculation of (P, P_M)

Recall

$$\Delta = (\mathrm{Gr}(\pi^\bullet) \times \{0\}) \sqcup (S \times \mathbb{R}^\times) \xrightarrow{q} \Sigma_{\mathrm{ad}}$$

and

$$1 \rightarrow M \times S^1 \rightarrow \Sigma \xrightarrow{p} S \rightarrow 1$$

Definition

$$\mathcal{Q} = q^{-1}(\mathcal{P}_{\mathrm{ad}}) \oplus (t^2 \Gamma_c(\ker dp))$$

Reeb component: $(t^2 \Gamma_c(\ker dp))$

Calculation

$$\mathcal{Q} = q^{-1} \left(\left(s_{\Sigma_{\mathrm{ad}}}^{-1} (pr_1^{-1} \mathcal{P}_M) \right) \oplus t (r_{\Sigma}^{-1} \mathcal{C}) \right) \oplus (t^2 \Gamma_c(\ker dp))$$

Analysing the osculation 1

$$\mathcal{Q} = q^{-1} \left(\left(s_{\Sigma_{\text{ad}}}^{-1} (\text{pr}_1^{-1} \mathcal{P}_M) \right) \oplus t \left(r_{\Sigma}^{-1} \mathcal{C} \right) \right) \oplus (t^2 \Gamma_c(\ker dp))$$

$$\Delta = (\text{Gr}(\pi^\bullet) \times \{0\}) \sqcup (S \times \mathbb{R}^\times) \xrightarrow{q} \Sigma_{\text{ad}}$$

Observations:

- ▶ $\mathcal{Q}$ does not include ∂_t . Hence, leaves of $(\Delta, \mathcal{Q})$ have **no** component in $\mathbb{R}$ -direction.
- ▶ So, leaves of $\mathcal{Q}$ stay in s -fibers of Δ .
- ▶ Family $\mathcal{Q} = \{\mathcal{Q}_t\}_{t \in \mathbb{R}}$ of foliations:

$$\mathcal{Q}_t \text{ foliation of } S, t \neq 0, \quad \mathcal{Q}_0 \text{ foliation of } \text{Gr}(\pi^\bullet) \rightarrow M$$

Analysing the osculation 2

$$\mathcal{Q} = q^{-1} \left(\left(s_{\Sigma_{\text{ad}}}^{-1} (\text{pr}_1^{-1} \mathcal{P}_M) \right) \oplus t \left(r_{\Sigma}^{-1} \mathcal{C} \right) \right) \oplus (t^2 \Gamma_c(\ker dp))$$

At $t = 0$ we have

$$\mathcal{Q}_0|_M = C \oplus P_M \oplus \mathbb{R}$$

Put $\tau : \text{gr}(\pi^\bullet) \rightarrow M$ the projection. Then

$$T\text{Gr}(\pi^\bullet) = \tau^! TM \oplus \text{gr}(\pi^\bullet)$$

Therefore:

- ▶ $q^{-1} \left(s_{\Sigma_{\text{ad}}}^{-1} (\text{pr}_1^{-1} \mathcal{P}_M) \right)$ contributes to $\Gamma_c(\tau^! TM)$.
- ▶ $q^{-1} \left(t r_{\Sigma}^{-1} \mathcal{C} \right) \oplus (t^2 \Gamma_c(\ker dp))$ contributes to $\Gamma_c(\text{gr}(\pi^\bullet))$ and is vertical.

Obstruction to pushing GQ down to (M, π)

$\mathcal{Q}_0$ **not** vertical in bundle $\text{Gr}(\pi^\bullet)$, because of component $q^{-1} \left(s_{\Sigma_{\text{ad}}}^{-1} (\text{pr}_1^{-1} \mathcal{P}_M) \right)$.

Example: Symplectic sphere

- ▶ Prequantizing bundle:

$$1 \rightarrow S^2 \times S^1 \rightarrow \frac{S^3 \times S^3}{S^1} \xrightarrow{\pi} S^2 \times S^2 \rightarrow 1$$

- ▶ Polarization by $\mathbb{T}^2$ -action:

$$\mathcal{P}_{S^2} = \langle \partial_\phi \rangle, \quad \mathcal{P} = \mathcal{C} \oplus \mathcal{P}_{S^2} = \langle (\partial_\phi, 0), (0, \partial_\phi) \rangle, \quad \mathcal{C} = \langle (\partial_\phi, 0) \rangle$$

- ▶ $\text{gr}(\omega_\bullet) = \frac{\text{TS}^2 \oplus \mathbb{R}_{S^2}}{\text{TS}^2} \oplus \text{TS}^2 \longrightarrow S^2$

$$\Delta = (\text{Gr}(\omega) \times \{0\}) \sqcup \left(\left(\frac{S^3 \times S^3}{S^1} \right) \times \mathbb{R}^\times \right) \implies S^2 \times \mathbb{R}$$

- ▶ Let R generator of $\Gamma_c(\ker dp)$. Then:

$$\Omega = \langle (0, \theta_\phi), (t\theta_\phi, 0), t^2 R \rangle$$

Dealing with obstruction

M sits in $H(Q)$, $H(P_M)$, S . Get double Lie groupoid

$$\begin{array}{ccccc}
 N_M^{H(Q)} & \rightrightarrows & N_M^{H(P_M)} & = & AS \oplus C_Q \oplus P_M \rightrightarrows P_M \\
 \downarrow \downarrow & & \downarrow \downarrow & & \downarrow \downarrow & & \downarrow \downarrow \\
 N_M^S & \rightrightarrows & M & & AS & \rightrightarrows & M
 \end{array}$$

Get filtration

$$\pi_{\bullet}^Q : \quad \underline{0} \subseteq T^*M \oplus C \oplus P_M \subset AS \oplus C_Q \oplus P_M$$

and

$$\begin{array}{ccc}
 Q_{\Delta} = \text{DNC}_{\pi_{\bullet}^Q}(H(Q), M) & \rightrightarrows & \text{DNC}(H(P_M), M) \\
 \downarrow \downarrow & & \downarrow \downarrow \\
 \Delta & \rightrightarrows & M \times \mathbb{R}
 \end{array}$$

$$\Gamma_c(AQ_{\Delta}) = \left(t(C \oplus P_M) + t^2((C \oplus \mathbb{R}_M) \oplus P_M) \right)$$